

```

Last login: Tue Jun 30 13:57:57 on ttys001
adityadave@Adityas-MacBook-Air-3 ~ % psql -d tpch-200mb
psql (18.4)
Type "help" for help.

tpch-200mb=# -- STEP 1: Cleanup
DROP VIEW IF EXISTS lineitem_secure CASCADE;
RESET ROLE;

-- STEP 2: Secure view, security_barrier = true
CREATE VIEW lineitem_secure WITH (security_barrier = true) AS
SELECT l.*
FROM lineitem l
WHERE EXISTS (
    SELECT 1 FROM orders o
    WHERE o.o_orderkey = l.l_orderkey
    AND o.o_orderpriority IN ('1-URGENT', '2-HIGH')
);

-- STEP 3: Grants
GRANT SELECT ON lineitem_secure TO alice;
GRANT SELECT ON orders TO alice;

-- STEP 4: Benchmark
SET ROLE alice;

EXPLAIN (ANALYZE, VERBOSE, BUFFERS)
SELECT sum(l_extendedprice * l_discount) AS revenue
FROM lineitem_secure
WHERE l_quantity::text ~ '[1-9]$';
DROP VIEW
RESET
CREATE VIEW
GRANT
GRANT
SET
WARNING: Log 1: Planner Input:
DETAIL: QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs true
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
  {RANGETBENTRY
    :alias <>
    :eref
      {ALIAS
        :aliasname lineitem_secure
        :colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenum" "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_lines" "l_status" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")
      }
    :rtekind 1
  }
  {SUBQUERY
    {QUERY
      :commandType 1
      :querySource 0
      :canSetTag true
      :utilityStmt <>
      :resultRelation 0
      :hasAggs false
      :hasWindowFuncs false
      :hasTargetSRFs false
      :hasSubLinks true
      :hasDistinctOn false
      :hasRecursive false
      :hasModifyingCTE false
      :hasForUpdate false
      :hasRowSecurity false
      :hasGroupRTE false
      :isReturn false
      :cteList <>
      :rtable (
        {RANGETBENTRY
          :alias
            {ALIAS
              :aliasname l
              :colnames <>
            }
          :eref
            {ALIAS
              :aliasname l
              :colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenum" "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_lines" "l_status" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")
            }
          :rtekind 0
          :relid 17440
          :inh true
          :relkind r
          :rellockmode 1
          :perminfoindex 1
          :tablesample <>
          :lateral false
          :inFromCl true
          :securityQuals <>
        }
      )
      :rteperminfos (
        {RTEPERMISSIONINFO
          :relid 17440
          :inh true
          :requiredPerms 2
          :checkAsUser 10
          :selectedCols (b 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23)
          :insertedCols (b)
          :updatedCols (b)
        }
      )
    }
  }
  {JOINTREE
    {FROMEXPR
      :fromlist (
        {RANGETBLREF
          :rtindex 1
        }
      )
    }
  }
  :quals
    {SUBLINK
      :subLinkType 0
      :subLinkId 0
      :testexpr <>
      :opName <>
      :subselect
        {QUERY
          :commandType 1
          :querySource 0
          :canSetTag true
          :utilityStmt <>
          :resultRelation 0
          :hasAggs false
          :hasWindowFuncs false
          :hasTargetSRFs false
          :hasSubLinks false
        }
      }
    }
  }

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:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
  {RANGETBENTRY
    :alias
      {ALIAS
        :aliasname o
        :colnames <>
      }
    :eref
      {ALIAS
        :aliasname o
        :colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_
          _totalprice" "o_orderdate" "o_orderpriority" "o_clerk"
          "o_shippriority" "o_comment")
      }
    :rtekind 0
    :relid 17428
    :inh true
    :relkind r
    :rellockmode 1
    :perminfoindex 1
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
  }
)
:rteperminfos (
  {RTEPERMISSIONINFO
    :relid 17428
    :inh true
    :requiredPerms 2
    :checkAsUser 10
    :selectedCols (b 8 13)
    :insertedCols (b)
    :updatedCols (b)
  }
)
:jointree
  {FROMEXPR
    :fromlist (
      {RANGETBLREF
        :rtindex 1
      }
    )
  }
:quals
  {BOOLEXP
    :boolop and
    :args (
      {OPEXP
        :opno 96
        :opfuncid 65
        :opresulttype 16
        :opretset false
        :opcollid 0
        :inputcollid 0
        :args (
          {VAR
            :varno 1
            :varattno 1
            :vartype 23
            :vartypmod -1
            :varcollid 0
            :varnullingrels (b)
            :varlevelsup 0
            :varreturningtype 0
            :varnosyn 1
            :varattnosyn 1
            :location -1
          }
          {VAR
            :varno 1
            :varattno 1
            :vartype 23
            :vartypmod -1
            :varcollid 0
            :varnullingrels (b)
            :varlevelsup 1
            :varreturningtype 0
            :varnosyn 1
            :varattnosyn 1
            :location -1
          }
        )
      }
      :location -1
    )
  }
  {SCALARARRAYOPEXP
    :opno 1054
    :opfuncid 1048
    :hashfuncid 0
    :negfuncid 0
    :useOr true
    :inputcollid 100
    :args (
      {VAR
        :varno 1
        :varattno 6
        :vartype 1042
        :vartypmod 19
        :varcollid 100
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 6
        :location -1
      }
      {ARRAYEXP
        :array_typeid 1014
        :array_collid 100
        :element_typeid 1042
        :elements (
          {CONST
            :consttype 1042
            :consttypmod -1
            :constcollid 100
            :constlen -1
            :constbyval false
            :constisnull false
            :location -1
            :constvalue 12 [ 48 0 0 0 49 45 85 82 71 69 7
              0 84 ]
          }
          {CONST
            :consttype 1042
            :consttypmod -1
            :constcollid 100
            :constlen -1
            :constbyval false
            :constisnull false
            :location -1
            :constvalue 10 [ 40 0 0 0 50 45 72 73 71 72 ]
          }
        )
      }
      :multidims false
      :list_start -1
      :list_end -1
      :location -1
    )
  }
  :location -1

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    }
    }
    :location -1
  }
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
    {CONST
      :consttype 23
      :consttypmod -1
      :constcollid 0
      :constlen 4
      :constbyval true
      :constisnull false
      :location -1
      :constvalue 4 [ 1 0 0 0 0 0 0 0 ]
    }
    :resno 1
    :resname ?column?
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
  }
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
    {VAR
      :varno 1
      :varattno 1
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 1
      :location -1
    }
    :resno 1
    :resname l_orderkey
    :ressortgroupref 0
    :resorigtbl 17440
    :resorigcol 1
    :resjunk false
  }
  {TARGETENTRY
    :expr
    {VAR
      :varno 1
      :varattno 2
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 2
      :location -1
    }
    :resno 2
    :resname l_partkey
    :ressortgroupref 0
    :resorigtbl 17440
    :resorigcol 2
    :resjunk false
  }
  {TARGETENTRY
    :expr
    {VAR
      :varno 1
      :varattno 3
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 3
      :location -1
    }
    :resno 3
    :resname l_suppkey
    :ressortgroupref 0
    :resorigtbl 17440
    :resorigcol 3
    :resjunk false
  }
  {TARGETENTRY
    :expr
    {VAR
      :varno 1
      :varattno 4
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 4
      :location -1
    }
    :resno 4
    :resname l_linenummer
    :ressortgroupref 0
    :resorigtbl 17440
    :resorigcol 4
    :resjunk false
  }
)
{TARGETENTRY
:expr

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{VAR
:varno 1
:varattno 5
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location -1
}
:resno 5
:resname l_quantity
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 5
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 6
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
}
:resno 6
:resname l_extendedprice
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 7
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 7
:location -1
}
}
:resno 7
:resname l_discount
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 7
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 8
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 8
:location -1
}
}
:resno 8
:resname l_tax
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 8
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 9
:vartype 1042
:vartypmod 5
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 9
:location -1
}
}
:resno 9
:resname l_returnflag
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 9
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 10
:vartype 1042
:vartypmod 5
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 10
:location -1
}
}
:resno 10
:resname l_linestatus
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 10
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 11
:vartype 1002
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 11
:location -1
}
}
:resno 11
:resname l_shipdate
:ressortgroupref 0

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:resorigtbl 17440
:resorigcol 11
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 12
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 12
:location -1
}
:resno 12
:resname l_commitdate
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 12
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 13
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 13
:location -1
}
:resno 13
:resname l_receiptdate
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 13
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 14
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 14
:location -1
}
:resno 14
:resname l_shipinstruct
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 14
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 15
:vartype 1042
:vartypmod 14
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 15
:location -1
}
:resno 15
:resname l_shipmode
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 15
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 16
:vartype 1043
:vartypmod 48
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 16
:location -1
}
:resno 16
:resname l_comment
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 16
:resjunk false
}
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:security_barrier true
:reldid 18515
:inh false
:relkind v
:rellockmode 1
:perminfoindex 1
:lateral false
:inFromCl true
:securityQuals <>
}
)

```

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:rteperminfos (
  {RTEPERMISSIONINFO
  :releid 18515
  :inh true
  :requiredPerms 2
  :checkAsUser 0
  :selectedCols (b 12 13 14)
  :insertedCols (b)
  :updatedCols (b)
  }
)
:jointree
{FROMEXPR
:fromlist (
  {RANGETBLREF
  :rtindex 1
  }
)
:quals
  {OEXPR
  :opno 641
  :opfuncid 1254
  :opresulttype 16
  :opretset false
  :opcollid 0
  :inputcollid 100
  :args (
    {COERCEVIAIO
    :arg
    {VAR
    :varno 1
    :varattno 5
    :vartype 1700
    :vartypmod 983046
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 5
    :location 115
    }
    :resulttype 25
    :resultcollid 100
    :coerceformat 1
    :location 125
    }
    {CONST
    :consttype 25
    :consttypmod -1
    :constcollid 100
    :constlen -1
    :constbyval false
    :constisnull false
    :location 134
    :constvalue 11 [ 44 0 0 0 94 91 49 45 57 93 36 ]
    }
  )
  :location 132
  }
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
  :expr
  {AGGREF
  :aggtfnoid 2114
  :aggtype 1700
  :aggcollid 0
  :inputcollid 0
  :aggtranstype 0
  :aggargtypes (o 1700)
  :aggdirectargs <>
  :args (
    {TARGETENTRY
    :expr
    {OEXPR
    :opno 1760
    :opfuncid 1726
    :opresulttype 1700
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR
      :varno 1
      :varattno 6
      :vartype 1700
      :vartypmod 983046
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 6
      :location 47
      }
      {VAR
      :varno 1
      :varattno 7
      :vartype 1700
      :vartypmod 983046
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 7
      :location 65
      }
    )
    :location 63
    }
  :resno 1
  :resname <>
  :ressortgroupref 0
  :resorigtbl 0
  :resorigcol 0
  :resjunk false
  }
)
:aggorder <>
:aggdistinct <>
:aggfilter <>
:aggstar false
:aggvariadic false
:aggkind n
:aggpresorted false
:agglevelsup 0
:aggssplit 0
:aggno -1
:aggtransno -1
:location 43
}
:resno 1
:resname revenue
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>

```

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:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}

WARNING: Log 2 :: After pull up At starting of Path Generation:
DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs true
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rttable (
{RANGETBLENTY
:alias <>
:eref
{ALIAS
:aliasname lineitem_secure
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenumber" "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_linesatus" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")
}
:rtkind 1
:subquery
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rttable (
{RANGETBLENTY
:alias
{ALIAS
:aliasname 1
:colnames <>
}
:eref
{ALIAS
:aliasname 1
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenumber" "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_linestatus" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")
}
:rtkind 0
:relid 17440
:inh true
:relkind r
:rellockmode 1
:permindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rtperminfos (
{RTEPERMISSIONINFO
:relid 17440
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals
{SUBLINK
:subLinkType 0
:subLinkId 0
:testexpr <>
:opName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rttable (
{RANGETBLENTY
:alias
{ALIAS
:aliasname 0
:colnames <>
}
:eref
{ALIAS

```

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        :aliasname o
        :colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_
          _totalprice" "o_orderdate" "o_orderpriority" "o_clerk"
            "o_shippriority" "o_comment")
        }
        :rtekind 0
        :relid 17428
        :inh true
        :relkind r
        :rellocmode 1
        :permindex 1
        :tablesample <>
        :lateral false
        :inFromCl true
        :securityQuals <>
        }
    )
    :rteperminfos (
        {RTEPERMISSIONINFO
        :relid 17428
        :inh true
        :requiredPerms 2
        :checkAsUser 10
        :selectedCols (b 8 13)
        :insertedCols (b)
        :updatedCols (b)
        }
    )
    :jointree
    {FROMEXPR
    :fromlist (
        {RANGETBLREF
        :rtindex 1
        }
    )
    :quals
    {BOOLEXP
    :boolop and
    :args (
        {OPEXP
        :opno 96
        :opfuncid 65
        :opresulttype 16
        :opretset false
        :opcollid 0
        :inputcollid 0
        :args (
            {VAR
            :varno 1
            :varattno 1
            :vartype 23
            :varitypmode -1
            :varcollid 0
            :varnullingrels (b)
            :varlevelsup 0
            :varreturningtype 0
            :varnosyn 1
            :varattnosyn 1
            :location -1
            }
            {VAR
            :varno 1
            :varattno 1
            :vartype 23
            :varitypmode -1
            :varcollid 0
            :varnullingrels (b)
            :varlevelsup 1
            :varreturningtype 0
            :varnosyn 1
            :varattnosyn 1
            :location -1
            }
        )
        :location -1
    }
    {SCALARARRAYOPEXP
    :opno 1054
    :opfuncid 1048
    :hashfuncid 0
    :negfuncid 0
    :useOr true
    :inputcollid 100
    :args (
        {VAR
        :varno 1
        :varattno 6
        :vartype 1042
        :varitypmode 19
        :varcollid 100
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 6
        :location -1
        }
        {ARRAYEXP
        :array_typeid 1014
        :array_collid 100
        :element_typeid 1042
        :elements (
            {CONST
            :consttype 1042
            :consttypmod -1
            :constcollid 100
            :constlen -1
            :constbyval false
            :constisnull false
            :location -1
            :constvalue 12 [ 48 0 0 0 49 45 85 82 71 69 7
              8 84 ]
            }
            {CONST
            :consttype 1042
            :consttypmod -1
            :constcollid 100
            :constlen -1
            :constbyval false
            :constisnull false
            :location -1
            :constvalue 10 [ 40 0 0 0 50 45 72 73 71 72 ]
            }
        )
        :multidims false
        :list_start -1
        :list_end -1
        :location -1
    }
    )
    :location -1
    }
    :location -1
    }
    }
    :mergeActionList <>
    :mergeTargetRelation 0
    :mergeJoinCondition <>
    :targetList (
        {TARGETENTRY
        :expr
        {CONST
        :consttype 23
        :consttypmod -1
        :constcollid 0
        :constlen 4
        :constbyval true

```

```

        :constisnull false
        :location -1
        :constvalue 4 [ 1 0 0 0 0 0 0 0 ]
    }
    :resno 1
    :resname ?column?
    :ressortgrouppref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
:resno 1
:resname l_orderkey
:ressortgrouppref 0
:resorigtbl 17440
:resorigcol 1
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
:resno 2
:resname l_partkey
:ressortgrouppref 0
:resorigtbl 17440
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
:resno 3
:resname l_suppkey
:ressortgrouppref 0
:resorigtbl 17440
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location -1
}
:resno 4
:resname l_linenumber
:ressortgrouppref 0
:resorigtbl 17440
:resorigcol 4
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 5
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location -1
}
:resno 5
:resname l_quantity
:ressortgrouppref 0
:resorigtbl 17440

```

```

:resorigcol 5
:resjunk false
}
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 6
:varstype 1700
:varstpmid 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
}
:resno 6
:resname l_extendedprice
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 6
:resjunk false
}
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 7
:varstype 1700
:varstpmid 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 7
:location -1
}
}
:resno 7
:resname l_discount
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 7
:resjunk false
}
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 8
:varstype 1700
:varstpmid 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 8
:location -1
}
}
:resno 8
:resname l_tax
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 8
:resjunk false
}
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 9
:varstype 1042
:varstpmid 5
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 9
:location -1
}
}
:resno 9
:resname l_returnflag
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 9
:resjunk false
}
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 10
:varstype 1042
:varstpmid 5
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 10
:location -1
}
}
:resno 10
:resname l_linestatus
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 10
:resjunk false
}
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 11
:varstype 1082
:varstpmid -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 11
:location -1
}
}
:resno 11
:resname l_shipdate
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 11
:resjunk false
}
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 12
:varstype 1082
:varstpmid -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 12

```

```

        :location -1
    }
    :resno 12
    :resname 1_commitdate
    :ressortgroupref 0
    :resorigtbl 17440
    :resorigcol 12
    :resjunk false
    }
    {TARGETENTRY
    :expr
    {VAR
    :varno 1
    :varattno 13
    :varatype 1082
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 13
    :location -1
    }
    :resno 13
    :resname 1_receiptdate
    :ressortgroupref 0
    :resorigtbl 17440
    :resorigcol 13
    :resjunk false
    }
    {TARGETENTRY
    :expr
    {VAR
    :varno 1
    :varattno 14
    :varatype 1042
    :vartypmod 29
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 14
    :location -1
    }
    :resno 14
    :resname 1_shipinstruct
    :ressortgroupref 0
    :resorigtbl 17440
    :resorigcol 14
    :resjunk false
    }
    {TARGETENTRY
    :expr
    {VAR
    :varno 1
    :varattno 15
    :varatype 1042
    :vartypmod 14
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 15
    :location -1
    }
    :resno 15
    :resname 1_shipmode
    :ressortgroupref 0
    :resorigtbl 17440
    :resorigcol 15
    :resjunk false
    }
    {TARGETENTRY
    :expr
    {VAR
    :varno 1
    :varattno 16
    :varatype 1043
    :vartypmod 48
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 16
    :location -1
    }
    :resno 16
    :resname 1_comment
    :ressortgroupref 0
    :resorigtbl 17440
    :resorigcol 16
    :resjunk false
    }
    }
    :override 0
    :onConflict <>
    :returningOldAlias <>
    :returningNewAlias <>
    :returningList <>
    :groupClause <>
    :groupDistinct false
    :groupingSets <>
    :havingQual <>
    :windowClause <>
    :distinctClause <>
    :sortClause <>
    :limitOffset <>
    :limitCount <>
    :limitOption 0
    :rowMarks <>
    :setOperations <>
    :constraintDecls <>
    :withCheckOptions <>
    :stmt_location -1
    :stmt_len -1
    }
    :security_barrier true
    :relid 18515
    :inh false
    :relkind v
    :rellockmode 1
    :perminfoindex 1
    :lateral false
    :inFromCl true
    :securityQuals <>
    }
    }
    :rtoperinfos (
    {RTEPERMISSIONINFO
    :relid 18515
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 12 13 14)
    :insertedCols (b)
    :updatedCols (b)
    }
    )
    :jointree
    {FROMEXPR
    :fromlist (
    {RANGETBLREF
    :rtindex 1
    }
    )
    }

```

```

)
:quals (
{OPEXPR
:opno 641
:opfuncid 1254
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{COERCEVIAIO
:arg
{VAR
:varno 1
:varattno 5
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 115
}
:resulttype 25
:resultcollid 100
:coerceformat 1
:location 125
}
{CONST
:consttype 25
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location 134
:constvalue 11 [ 44 0 0 0 94 91 49 45 57 93 36 ]
}
)
:location 132
}
)
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{AGGREF
:aggfnoid 2114
:aggtype 1700
:aggcollid 0
:inputcollid 0
:aggtranstype 0
:aggargtypes (o 1700)
:aggdirectargs <>
:args (
{TARGETENTRY
:expr
{OPEXPR
:opno 1760
:opfuncid 1726
:opresulttype 1700
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 6
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location 47
}
{VAR
:varno 1
:varattno 7
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 7
:location 65
}
)
:location 63
}
:resno 1
:resname <>
:ressortgroupref 0
:ressorttbl 0
:resorigcol 0
:resjunk false
}
)
:aggorder <>
:aggdistinct <>
:aggfilter <>
:aggstar false
:aggvariadic false
:aggkind n
:aggpresorted false
:agglevelsup 0
:aggsplit 0
:aggno -1
:aggtransno -1
:location 43
}
:resno 1
:resname revenue
:ressortgroupref 0
:ressorttbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0

```

```

:stmt_len 0
}

WARNING: is_parallel_safe: INPUT qual = ((RESTRICTINFO :clause {OEXPR :opno 641 :opfuncid 1254 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ((COERCEVIAIO :arg {VAR :varno 1 :var
attno 5 :vartype 1700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 5 :location -1} :resulttype 25 :resultcollid 100 :coerceformat 1 :loc
ation -1} {CONST :consttype 25 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 11 [ 44 0 0 0 94 91 49 45 57 93 36 ]}) :location -1} :is_pushed
_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompa
tible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left
_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})
WARNING: is_parallel_safe: OUTPUT qual = ((RESTRICTINFO :clause {OEXPR :opno 641 :opfuncid 1254 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ((COERCEVIAIO :arg {VAR :varno 1 :va
rattno 5 :vartype 1700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 5 :location -1} :resulttype 25 :resultcollid 100 :coerceformat 1 :lo
cation -1} {CONST :consttype 25 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 11 [ 44 0 0 0 94 91 49 45 57 93 36 ]}) :location -1} :is_pushe
d_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incomp
atible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :lef
t_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | parallel_safe
= true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ((VAR :varno 1 :varattno 6 :vartype 1700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :locati
on -1} {VAR :varno 1 :varattno 7 :vartype 1700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 7 :location -1})
WARNING: is_parallel_safe: OUTPUT qual = ((VAR :varno 1 :varattno 6 :vartype 1700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :locat
ion -1} {VAR :varno 1 :varattno 7 :vartype 1700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 7 :location -1}) | parallel_safe = true | r
eason = PROPARALLEL_SAFE
WARNING: Log 2 : After pull up At starting of Path Generation:
DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctList <>
:table (
{RANGETBENTRY
:alias
{ALIAS
:aliasname 1
:colnames <>
}
:eref
{ALIAS
:aliasname 1
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenummer" "l_qua
ntity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_lines
tatus" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "
l_shipmode" "l_comment")
}
:rtekind 0
:relid 17440
:inh false
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias
{ALIAS
:aliasname o
:colnames <>
}
:eref
{ALIAS
:aliasname o
:colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o
_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
}
:rtekind 0
:relid 17428
:inh false
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 17440
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17428
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 13)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{JOINEXPR
:jointype 4
:isNatural false
:larg
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals <>
}
}
:rarg
{RANGETBLREF
:rtindex 2
}
)
:usingClause <>
:join_using_alias <>
:quals (
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0

```

```

:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location -1
}
{VAR
:varno 1
:varattno 1
:varatype 23
:varatypepm -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
)
:location -1
}
{SCALARARRAYOEXPR
:opno 1054
:opfuncid 1048
:hashfuncid 0
:negfuncid 0
:useOr true
:inputcollid 100
:args (
{VAR
:varno 2
:varattno 6
:varatype 1042
:varatypepm 19
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 6
:location -1
}
{CONST
:consttype 1014
:consttypepm -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 48 [-64 0 0 0 1 0 0 0 0 0 0 0 18 4 0 0 2 0 0 0 1 0
0 0 48 0 0 0 49 45 85 82 71 69 78 84 40 0 0 0 50 45 72 73 71 7
2 0 0 0 ]
}
)
:location -1
}
)
:alias <>
:rtindex 0
}
)
:quals <>
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{CONST
:consttype 23
:consttypepm -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull true
:location -1
:constvalue <>
}
:resno 1
:resname l_orderkey
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 1
:resjunk false
}
{TARGETENTRY
:expr
{CONST
:consttype 23
:consttypepm -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull true
:location -1
:constvalue <>
}
:resno 2
:resname l_partkey
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{CONST
:consttype 23
:consttypepm -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull true
:location -1
:constvalue <>
}
:resno 3
:resname l_supkey
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{CONST
:consttype 23
:consttypepm -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull true
:location -1
:constvalue <>
}
:resno 4
:resname l_linenum
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 4
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1

```

```

:varattno 5
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location -1
}
:resno 5
:resname l_quantity
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 5
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 6
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
:resno 6
:resname l_extendedprice
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 7
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 7
:location -1
}
:resno 7
:resname l_discount
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 7
:resjunk false
}
{TARGETENTRY
:expr
{CONST
:consttype 1700
:consttypmod 983046
:constcollid 0
:constlen -1
:constbyval false
:constisnull true
:location -1
:constvalue <>
}
:resno 8
:resname l_tax
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 8
:resjunk false
}
{TARGETENTRY
:expr
{CONST
:consttype 1042
:consttypmod 5
:constcollid 100
:constlen -1
:constbyval false
:constisnull true
:location -1
:constvalue <>
}
:resno 9
:resname l_returnflag
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 9
:resjunk false
}
{TARGETENTRY
:expr
{CONST
:consttype 1042
:consttypmod 5
:constcollid 100
:constlen -1
:constbyval false
:constisnull true
:location -1
:constvalue <>
}
:resno 10
:resname l_linestatus
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 10
:resjunk false
}
{TARGETENTRY
:expr
{CONST
:consttype 1082
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull true
:location -1
:constvalue <>
}
:resno 11
:resname l_shipdate
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 11
:resjunk false
}
{TARGETENTRY
:expr
{CONST
:consttype 1082
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull true
:location -1
}

```

```

        :constvalue <>
    }
    :resno 12
    :resname l_commitdate
    :resortgroupref 0
    :resorigtbl 17440
    :resorigcol 12
    :resjunk false
    }
    {TARGETENTRY
    :expr
    {CONST
    :consttype 1082
    :consttypmod -1
    :constcollid 0
    :constlen 4
    :constbyval true
    :constisnull true
    :location -1
    :constvalue <>
    }
    :resno 13
    :resname l_receiptdate
    :resortgroupref 0
    :resorigtbl 17440
    :resorigcol 13
    :resjunk false
    }
    {TARGETENTRY
    :expr
    {CONST
    :consttype 1042
    :consttypmod 29
    :constcollid 100
    :constlen -1
    :constbyval false
    :constisnull true
    :location -1
    :constvalue <>
    }
    :resno 14
    :resname l_shipinstruct
    :resortgroupref 0
    :resorigtbl 17440
    :resorigcol 14
    :resjunk false
    }
    {TARGETENTRY
    :expr
    {CONST
    :consttype 1042
    :consttypmod 14
    :constcollid 100
    :constlen -1
    :constbyval false
    :constisnull true
    :location -1
    :constvalue <>
    }
    :resno 15
    :resname l_shipmode
    :resortgroupref 0
    :resorigtbl 17440
    :resorigcol 15
    :resjunk false
    }
    {TARGETENTRY
    :expr
    {CONST
    :consttype 1043
    :consttypmod 48
    :constcollid 100
    :constlen -1
    :constbyval false
    :constisnull true
    :location -1
    :constvalue <>
    }
    :resno 16
    :resname l_comment
    :resortgroupref 0
    :resorigtbl 17440
    :resorigcol 16
    :resjunk false
    }
    }
    :override 0
    :onConflict <>
    :returningOldAlias <>
    :returningNewAlias <>
    :returningList <>
    :groupClause <>
    :groupDistinct false
    :groupingSets <>
    :havingQual <>
    :windowClause <>
    :distinctClause <>
    :sortClause <>
    :limitOffset <>
    :limitCount <>
    :limitOption 0
    :rowMarks <>
    :setOperations <>
    :constraintDepds <>
    :withCheckOptions <>
    :stmt_location -1
    :stmt_len -1
    }
)
WARNING:  is_parallel_safe: INPUT qual = <>
WARNING:  is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING:  is_parallel_safe: INPUT qual = ({VAR :varno 1 :varattno 5 :vartype 1700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 5 :locati
on -1} {VAR :varno 1 :varattno 6 :vartype 1700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1} {VAR :varno 1 :varattno 7 :v
artype 1700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 7 :location -1} {VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid
0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1})
WARNING:  is_parallel_safe: OUTPUT qual = ({VAR :varno 1 :varattno 5 :vartype 1700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 5 :locat
ion -1} {VAR :varno 1 :varattno 6 :vartype 1700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1} {VAR :varno 1 :varattno 7 :v
artype 1700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 7 :location -1} {VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid
0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} {CONST :consttype 1014 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 48 [-64 0 0 0 1 0 0 0 0 0 0 0 18 4 0 0 2 0 0 0 1 0 0 0 48 0 0 0 49 45 85 82 71 69 78 84 40 0 0 0 50 45 72 73 71 72 0 0 ]} :location -1}
:is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 2) :required_relds (b 2
) :incompatible_relds (b) :outer_relds (b) :left_relds (b) :right_relds (b) :orclause <> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies
<> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})
WARNING:  is_parallel_safe: OUTPUT qual = ({RESTRICTINFO :clause {SCALARARRAYOPEXPR :opno 1054 :opfuncid 1048 :hashfuncid 0 :negfuncid 0 :useOr true :inputcollid 100 :args ({VAR :varno 2 :varattno 6 :var
type 1042 :vartypmod 19 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1} {CONST :consttype 1014 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 48 [-64 0 0 0 1 0 0 0 0 0 0 0 18 4 0 0 2 0 0 0 1 0 0 0 48 0 0 0 49 45 85 82 71 69 78 84 40 0 0 0 50 45 72 73 71 72 0 0 ]} :location -1}
:is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 2) :required_relds (b 2
) :incompatible_relds (b) :outer_relds (b) :left_relds (b) :right_relds (b) :orclause <> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies
<> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | paral
lel_safe = true | reason = PROPARALLEL_SAFE
WARNING:  is_parallel_safe: INPUT qual = ({VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1}
)
WARNING:  is_parallel_safe: OUTPUT qual = ({VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1
}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING:  is_parallel_safe: INPUT qual = ({RESTRICTINFO :clause {OPEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 1 :vartype 23 :var
typmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} {VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (
b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1} :location -1} :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof false
:has_volatile 2 :security_level 0 :num_base_rels 2 :clause_relds (b 1 2) :required_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b 2) :orclause <> :rinfo_s
erial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec 0.39916666666666667 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 1 :vartype 23
:vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdom
ain {JOINDOMAIN :jd_relds (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnos
yn 2 :varattnosyn 1 :location -1} :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 96 :left_buck
etsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 96 :right_hasheqoperator 96})
WARNING:  is_parallel_safe: OUTPUT qual = ({RESTRICTINFO :clause {OPEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 1 :vartype 23 :v
artypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} {VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels
(b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1} :location -1} :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof false

```

QUERY PLAN

```
Finalize Aggregate (cost=43085.58..43085.59 rows=1 width=32) (actual time=280.433..282.704 rows=1.00 loops=1)
  Output: sum((lineitem_secure.l_extendedprice * lineitem_secure.l_discount))
  Buffers: shared hit=14366 read=13458
  -> Gather (cost=43085.35..43085.56 rows=2 width=32) (actual time=280.391..282.688 rows=3.00 loops=1)
    Output: (PARTIAL sum((lineitem_secure.l_extendedprice * lineitem_secure.l_discount)))
    Workers Planned: 2
    Workers Launched: 2
    Buffers: shared hit=14366 read=13458
  -> Partial Aggregate (cost=42085.35..42085.36 rows=1 width=32) (actual time=275.493..275.497 rows=1.00 loops=3)
    Output: PARTIAL sum((lineitem_secure.l_extendedprice * lineitem_secure.l_discount))
    Buffers: shared hit=14366 read=13458
    Worker 0: actual time=273.158..273.160 rows=1.00 loops=1
      Buffers: shared hit=4775 read=4443
    Worker 1: actual time=273.153..273.155 rows=1.00 loops=1
      Buffers: shared hit=4719 read=4430
  -> Subquery Scan on lineitem_secure (cost=7458.20..42035.46 rows=9979 width=12) (actual time=275.488..275.491 rows=0.00 loops=3)
    Output: lineitem_secure.l_orderkey, lineitem_secure.l_partkey, lineitem_secure.l_supkey, lineitem_secure.l_linenum, lineitem_secure.l_extendedprice,
lineitem_secure.l_discount, lineitem_secure.l_tax, lineitem_secure.l_returnflag, lineitem_secure.l_linestatus, lineitem_secure.l_shipdate, lineitem_secure.l_commitdate, li
neitem_secure.l_shipinstruct, lineitem_secure.l_shipmode, lineitem_secure.l_comment
    Filter: ((lineitem_secure.l_quantity)::text ~ '^[1-9]$')::text
    Rows Removed by Filter: 160171
    Buffers: shared hit=14366 read=13458
    Worker 0: actual time=273.155..273.157 rows=0.00 loops=1
      Buffers: shared hit=4775 read=4443
    Worker 1: actual time=273.149..273.151 rows=0.00 loops=1
      Buffers: shared hit=4719 read=4430
  -> Parallel Hash Semi Join (cost=7458.20..38542.84 rows=199578 width=333) (actual time=37.983..202.111 rows=160171.00 loops=3)
    Output: NULL::integer, NULL::integer, NULL::integer, NULL::integer, 1.1_quantity, 1.1_extendedprice, 1.1_discount, NULL::numeric(15,2), NULL::character(1), NULL::character(1),
NULL::date, NULL::date, NULL::date, NULL::character(25), NULL::character(10), NULL::character varying(44)
    Hash Cond: (1.1_orderkey = o.o_orderkey)
    Buffers: shared hit=14366 read=13458
    Worker 0: actual time=35.768..199.775 rows=160794.00 loops=1
      Buffers: shared hit=4775 read=4443
```

```
Worker 1: actual time=36.083..199.568 rows=160430.00 loops=1
  Buffers: shared hit=4719 read=4430
-> Parallel Seq Scan on public.lineitem l (cost=0.00..27551.87 rows=499987 width=21) (actual time=0.630..50.317 rows=399989.67 loops=3)
  Output: l.l_orderkey, l.l_partkey, l.l_suppkey, l.l_linenum, l.l_quantity, l.l_extendedprice, l.l_discount, l.l_tax, l.l_returnflag, l.l_linestatus, l.l_shipdate, l.l_
_commitdate, l.l_receiptdate, l.l_shipinstruct, l.l_shipmode, l.l_comment
  Buffers: shared hit=9934 read=12618
Worker 0: actual time=0.577..50.289 rows=402091.00 loops=1
  Buffers: shared hit=3331 read=4226
Worker 1: actual time=0.899..49.849 rows=397790.00 loops=1
  Buffers: shared hit=3299 read=4176
-> Parallel Hash (cost=6034.50..6834.50 rows=49896 width=4) (actual time=37.152..37.153 rows=40048.00 loops=3)
  Output: o.o_orderkey
  Buckets: 131072 Batches: 1 Memory Usage: 5760kB
  Buffers: shared hit=4432 read=840
Worker 0: actual time=35.153..35.153 rows=38039.00 loops=1
  Buffers: shared hit=1444 read=217
Worker 1: actual time=35.149..35.150 rows=38020.00 loops=1
  Buffers: shared hit=1420 read=254
-> Parallel Seq Scan on public.orders o (cost=0.00..6834.50 rows=49896 width=4) (actual time=0.086..30.674 rows=40048.00 loops=3)
  Output: o.o_orderkey
  Filter: (o.o_orderpriority = ANY ('{1-URGENT,2-HIGH}'))::bpchar[])
  Rows Removed by Filter: 59952
  Buffers: shared hit=4432 read=840
Worker 0: actual time=0.015..28.876 rows=38039.00 loops=1
  Buffers: shared hit=1444 read=217
Worker 1: actual time=0.011..29.307 rows=38020.00 loops=1
  Buffers: shared hit=1420 read=254

Planning:
  Buffers: shared hit=198 read=2
Planning Time: 8.867 ms
Execution Time: 283.140 ms
(60 rows)

tpch-200mb=>
```